TechFusion — Round 2 Build Submission

Brand Trust, Before Publish · Intelligent Brand-Governance Platform

Problem Statement: Brand Governance (Inkloom AI)	**Round:** Round 2 — The Build (100-Point Evaluation)
Team Name: TechFusion	**Date:** October 26, 2026
Team Members: • Yash Singla (Leader — Product & Integration) • Rishit Negi (Review-Engine Logic) • Ramandeep Singh (Platform & Workflow) • Aditya Sharma (UX, Testing & Demo Readiness)	**Submission Artifacts:** • Live Deployed System: Render Cloud & Local Port 8000 • Source Repository: https://github.com/furydarkknight/techfusion-brand-governance • Audit Engine: TechFusion Guardrail v2.4

1. Core Workflow & End-to-End Execution (30 Points)

TechFusion is an explainable brand-governance system built to eliminate preventable reputation risk before publication. Unlike aggressive generative rewrites that erase human voice, TechFusion operates as a *transparent guardrail*. The end-to-end workflow functions reliably from content ingestion to managerial sign-off:

- **Ingest & Analyze:** Marketing or technical copy is ingested through the Live Studio or API. Content is parsed into atomic semantic tokens.
- **Multi-Dimensional Inspection:** The engine simultaneously evaluates 5 pillars: *Voice & Tone*, *Approved Terminology*, *Fact & SLA Verification*, *Visual & Color Palette Alignment*, and *WCAG 2.1 AA Accessibility*.
- **Evidence-Linked Flagging:** Every flagged phrase maps strictly to an authoritative guideline ID (e.g., GL-CLAIM-01) or audited fact record (e.g., FACT-01), complete with exact character offsets.
- **Purpose-Aware 1-Click Corrections:** Authors can accept proposed corrections that preserve intent or make manual adjustments with live scoring feedback.
- **Role-Based Collaboration:** Seamless switching between Creator, Brand Manager, and Authorised Reviewer roles with threaded discussion, revision snapshots, and auditable override reasons.
- **Audit Trail & Certificate Export:** Every decision, comment, and override is permanently committed to an immutable log with cryptographic timestamping.

2. List of Completed Features in Round 2

Module	Completed Capability	Status
Guideline Hub	Centralized repository for versioned voice, tone, terminology, visual guidelines, and WCAG accessibility standards with live add/edit capabilities.	**100% Complete**
Approved Facts DB	Audited claim repository linking claims (customer metrics, certifications, SLAs, encryption protocols) to source evidence documents.	**100% Complete**
Inspection Engine	Real-time regex and semantic scanner detecting tone drift, unverified superlatives, forbidden terminology, and reading ease.	**100% Complete**

Purpose-Aware Diff	Side-by-side visual diff showing original draft versus corrected proposal with 1-click 'Apply All' or individual item correction.	**100% Complete**
Multi-Role Workflow	Role switching across Creator, Brand Manager, and Authorised Reviewer with approval, revision request, and rationale-backed override.	**100% Complete**
Governance Insights	Analytics dashboard tracking recurring issues, department compliance heatmaps, drift trends, and guideline clarity alerts.	**100% Complete**
Audit & Export	Full tamper-evident audit logging and automated PDF Brand Governance Certificate generation.	**100% Complete**

3. Technology and Implementation Details (25 Points)

Architecture: TechFusion is designed with a decoupled, high-performance architecture optimized for sub-15ms audit latency:

• **Backend Framework:** Python 3.12 with FastAPI & Starlette, providing high-concurrency async endpoints, strict Pydantic schemas, and instant JSON serialization.

• **Review Engine Core:** Rule-based lexical tokenizer combined with contextual regex heuristics and statistical Flesch-Kincaid NLP analyzers to prevent non-deterministic hallucination.

• **Knowledge Layer:** Normalized Guideline & Verified Fact database schema with versioning (v1.0 -> v3.0) and citation mapping.

• **Frontend Architecture:** Single Page Application utilizing HTML5, modern reactive JavaScript (ES6+), Tailwind CSS CDN, and dynamic Lucide SVG UI components.

• **Responsible AI Transparency:** Explicit disclosure badges detailing engine version, confidence thresholds, and audited source citations for every flagged item.

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4. Progress from Round 1 & Explanation of Plan Changes (20 Points)

In Round 1, our proposal outlined three phases: (1) Research & Schema, (2) Core Checks & Suggestions, and (3) Roles & Audit Trail. During Round 2 implementation, we validated that all three planned phases were successfully materialized into a functional product. Furthermore, based on user workflow testing, we introduced two crucial enhancements:

- **Evolution from Passive Warnings to Interactive Diff proposals:** Originally, we planned to present findings as a static warning list. We upgraded this into an interactive split-view diff generator where creators can inspect the exact purpose-preserved rewrite and apply corrections with one click.
- **Introduction of Mandatory Override Rationale:** We recognized that enterprise brand managers occasionally need to approve strategic exceptions (e.g., temporary promotional slogans). We implemented a governance override system that forces reviewers to log their formal justification, maintaining an auditable paper trail.
- **Zero-Latency In-Browser Verification:** Instead of sending confidential draft copy to external third-party cloud APIs, our engine performs local deterministic checks, safeguarding enterprise secrets.

5. Three Major Improvements Planned for Round 3 (15 Points)

The following three substantial, meaningful product expansions are scheduled for Round 3 'The Finish':

Improvement 1: Multi-Modal Computer Vision Brand Inspector

Expand beyond text and color-hex checks into automated computer vision inspection for marketing banners, PDFs, and slide decks. The engine will run automated OCR, logo safe-zone measurement, brand font hierarchy verification, and WCAG 2.1 AA foreground-to-background contrast ratio calculation directly on uploaded image files (PNG/JPEG/SVG/PDF).

Improvement 2: Vector-Grounded Semantic RAG & Dynamic Brand Guidelines Ingestion

Implement a dynamic vector embeddings pipeline (using ChromaDB/FAISS) that allows enterprise admins to upload raw PDF brand books, annual SEC 10-K filings, and press releases. The system will automatically parse, chunk, and index verified facts and tone rules with semantic similarity search, providing context-aware explanations for nuanced industry jargon.

Improvement 3: Real-Time CMS & Browser Extensions (Figma, Google Docs, Chrome)

Deliver real-time inline checking extensions for Google Docs, WordPress, and Figma. Creators will receive non-intrusive brand guardrail prompts while drafting directly inside their native design and authoring tools, eliminating the friction of manual copy-pasting.

6. Usability & 60-Second Demo Guide for Judges (10 Points)

Judges can verify the entire product workflow in under 60 seconds without technical setup:

Step 1: Open http://127.0.0.1:8000 in any web browser.

Step 2: In the **Live Review Studio**, click *'Load Risky Launch Announcement'* demo preset and click **'Run Brand Audit'**. Observe the multi-category score drop to 42/100, displaying 8 flagged issues with verified citations.

Step 3: Click **'Apply All Purpose-Preserved Corrections'**. Watch the text transform automatically into compliant enterprise copy and the score rise to 98/100.

Step 4: Switch user role from *Creator* to *Brand Reviewer* in the top bar. Add an official sign-off comment and click **'Approve & Certify'**.

Step 5: Explore the **Guideline Hub** (add or edit rules), **Governance Insights** (view recurring error distributions), and click **'Download Submission PDF'**.

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